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## **PYTHON BASICS ON TRY HACK ME**

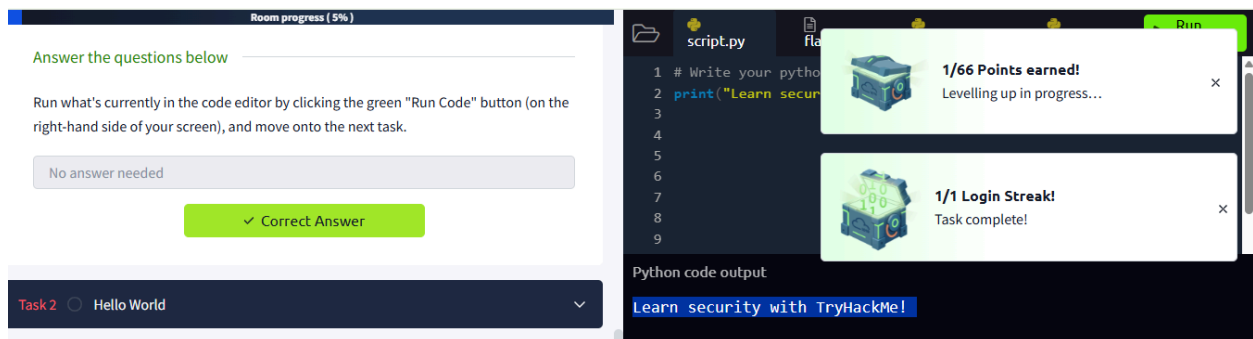
### **1.1 Introduction**

This module has provided beginners with basic knowledge of the Python programming language. I now understand how countdowns, such as those that happen on the 31st, work by counting downward using a for loop together with the time module. I also learned that many calculations can be executed using simple code, and that scripting in Python can be used for automation tasks.

### **1.2 Introduction to python**

Run what's currently in the code editor by clicking the green "Run Code" button (on the right-hand side of your screen), and move onto the next task.

Explanation: I was expected to understand that after writing a code, i get the output by run the command.



### **2.1 Hello World**

On the code editor, print "Hello World". What is the flag?

Answer: THM{PRINT\_STATEMENTS}

Room progress ( 11% )

```
# This is an example comment
print("Hello World")
```

As you can see from the example code block above, it's just one line (shown on line 2), and when we run this code, it will output the text `Hello World`. Let's break this down. In the example, line 1 is a comment, a line starting with a hashtag (#) symbol and is not run by the computer. A comment is written by the programmer (you) to help other people reading the code understand what is going on.

We can control what is output to the screen by using the `print()` statement. Anything inside of the parenthesis `()` will be output. However, because we are printing a string (more on data types later in this room), we have to put them inside of quotations `" "`.

Please note, this room's examples are for Python3.

Answer the questions below

On the code editor, print "Hello World". What is the flag?

THM{PRINT\_STATEMENTS}

script.py flag.txt

```
1 # Write your python code here
2 print("Hello World")
3
4
5
6
7
8
9
```

1/66 Points earned! Levelling up in progress...

8/66 Points Earned! Levelling up in progress...

Exercise Complete! The flag is: THM{PRINT\_STATEMENTS}

Python code output

Hello World

### 3.1 Mathematical Operations

a) In the code editor, print the result of  $21 + 43$ . What is the flag?

Answer: THM{ADDITION}

Room progress ( 16% )

|                          |    |
|--------------------------|----|
| Greater than or equal to | >= |
| Less than or equal       | <= |

Answer the questions below

In the code editor, print the result of  $21 + 43$ . What is the flag?

THM{ADDITION}

✓ Correct Answer

script.py flag.txt shipping.py bitcoin.py

```
1 # Write your python code here
2 print(21+43)
3
4
5
6
7
8
9
```

Exercise Complete! The flag is: THM{ADDITION}

Python code output

64

b) Print the result of  $142 - 52$ . What is the flag?

Answer: THM{SUBTRACT}

Room progress ( 22% )

Print the result of  $142 - 52$ . What is the flag?

THM{SUBTRACT}

✓ Correct Answer

Print the result of  $10 * 342$ . What is the flag?

{ }

Check

Print the result of 5 squared. What is the flag?

{ }

script.py flag.txt

```
1 # Write your python code here
2 print(142-52)
3
4
5
6
7
8
9
```

17/66 Points earned! Levelling up in progress...

24/66 Points Earned! Levelling up in progress...

Exercise Complete! The flag is: THM{SUBTRACT}

Python code output

90

c) Print the result of  $10 * 342$ . What is the flag?  
Answer: THM{MULTIPLICATION\_PYTHON}

The screenshot shows a coding challenge interface. On the left, a text input field contains the answer 'THM{MULTIPLICATION\_PYTHON}', which has been marked as 'Correct Answer'. Below it, another question asks for the result of 5 squared. On the right, a code editor shows a Python script: 

```
1 # Write your python code here
2 print(10*342)
3
4
5
6
7
8
9
```

 The 'Run Code' button is highlighted. Below the code editor, a message states 'Exercise Complete! The flag is: THM{MULTIPLICATION\_PYTHON}'. The Python code output is shown as '3420'.

d) Print the result of 5 squared. What is the flag?  
Answer: THM{EXPON3NT\_POWER}

The screenshot shows a coding challenge interface. On the left, a text input field contains the answer 'THM{EXPON3NT\_POWER}', which has been marked as 'Correct Answer'. Below it, another question asks for the result of 5 squared. On the right, a code editor shows a Python script: 

```
1 # Write your python code here
2 print(5**2)
3
4
5
6
7
8
9
```

 The 'Run Code' button is highlighted. Below the code editor, a message states 'Exercise Complete! The flag is: THM{EXPON3NT\_POWER}'. The Python code output is shown as '25'.

## 4.1 Variables and data types.

a) In the code editor, create a variable called height and set its initial value to 200.

Answer:

- **Integer** - Whole numbers
- **Float** - Numbers that contain decimal points or for fractions
- **Boolean** - Used for data that is restricted to True or False options
- **List** - Series of different data types stored in a collection

| <div>String   Float   Integer   Boolean   List</div> |        |              |          |              |
|------------------------------------------------------|--------|--------------|----------|--------------|
| Title                                                | Rating | Times Viewed | Favorite | Seen By      |
| Star Wars                                            | 9.8    | 13           | True     | Alice, Bob   |
| Matrix                                               | 8.5    | 23           | False    | Charlie      |
| Indiana Jones                                        | 6.1    | 3            | False    | Daniel, Evie |

Answer the questions below

In the code editor, create a variable called height and set its initial value to 200.

No answer needed

✓ Correct Answer

The screenshot shows a code editor with a file explorer at the top containing 'script.py', 'flag.txt', 'shipping.py', and 'bitcoin.py'. The 'script.py' file is open, showing the following code: 

```
1 height=200
2
3
4
5
6
7
8
9
```

 The 'Run Code' button is highlighted. Below the code editor, the 'Python code output' section is visible but empty.

b) On a new line, add 50 to the height variable.

Answer :

On a new line, add 50 to the height variable.

No answer needed

✓ Correct Answer

On another new line, print out the value of height. What is the flag that appears?

script.pyflag.txtshipping.pybitcoin.pyRun Code

```
1 height=200
2 height=height+50
3
4
5
6
7
8
```

c) On another new line, print out the value of height. What is the flag that appears?

Answer: THM{VARIABLE3S}

On another new line, print out the value of height. What is the flag that appears?

THM{VARIABLE3S}

✓ Correct Answer

script.pyflag.txtshipping.pybitcoin.pyRun Code

```
1 height=200
2 height=height+50
3 print(height)
4
5
6
7
8
9
```

Exercise Complete! The flag is: THM{VARIABLE3S}

Python code output

250

## 5.1 Logical and Boolean Operators

I needed to understand everything about if, else and elif.

```
a = 1
if a == 1 or a > 10:
    print("a is either 1 or above 10")
```

```
name = "bob"
hungry = True
if name == "bob" and hungry == True:
    print("bob is hungry")
elif name == "bob" and not hungry:
    print("Bob is not hungry")
else: # If all other if conditions are not met
    print("Not sure who this is or if they are hungry")
```

Answer the questions below

Read the above section.

No answer needed

✓ Correct Answer

## 6.1 Introduction to if statement

a) In the code editor, click on the "shipping.py" tab and follow the instructions to complete this task.

If statements are essential in programming and will be something you use a lot.

Answer the questions below

In this exercise, we will code a small application that calculates and outputs the shipping cost for a customer based on how much they've spent.

In the code editor, click on the "shipping.py" tab and follow the instructions to complete this task.

No answer needed

✓ Correct Answer

```
script.py  flag.txt  shipping.py  bitcoin.py  Code
14 shipping_cost_per_kg = 1.20
15 customer_basket_cost = 34
16 customer_basket_weight = 44
17
18 if(customer_basket_cost >= 34):
19     print('Free shipping!')
20 else:
21     shipping_cost = customer_basket_weight * shipping_cost_per_kg
22     customer_basket_cost = 34

Python code output
Free shipping!
Total basket cost including shipping is 34
```

b) Once you've written the application in the code editor's shipping.py tab, a flag will appear, which is the answer to this question.

Answer:THM{IF\_STATEMENT\_SHOPPING}

Once you've written the application in the code editor's shipping.py tab, a flag will appear, which is the answer to this question.

THM{IF\_STATEMENT\_SHOPPING}

✓ Correct Answer

In shipping.py, on line 15 (when using the Code Editor's Hint), change the **customer\_basket\_cost** variable to **101** and re-run your code. You will get a flag (if the total cost is correct based on your code); the flag is the answer to this question.

\_\_\_\_{\_\_\_\_\_}

```
16 customer_basket_weight = 44
17
18 if(customer_basket_cost >= 100):
19     print('Free shipping!')
20 else:
21     shipping_cost = customer_basket_weight * shipping_cost_per_kg
22     customer_basket_cost = 86.8
23
24 print("Total basket cost including shipping is " +
25       str(customer_basket_cost))
```

Exercise Complete! The flag is: THM{IF\_STATEMENT\_SHOPPING}

Python code output

Total basket cost including shipping is 86.8

c)In shipping.py, on line 15 (when using the Code Editor's Hint), change the **customer\_basket\_cost** variable to **101** and re-run your code. You will get a flag (if the total cost is correct based on your code); the flag is the answer to this question.

Answer:THM{MY\_FIRST\_APP}

In shipping.py, on line 15 (when using the Code Editor's Hint), change the **customer\_basket\_cost** variable to **101** and re-run your code. You will get a flag (if the total cost is correct based on your code); the flag is the answer to this question.

THM{MY\_FIRST\_APP}

✓ Correct Answer

Task 7 ○ Loops

```
14 shipping_cost_per_kg = 1.20
15 customer_basket_cost = 101
16 customer_basket_weight = 44
17
18 if(customer_basket_cost >= 100):
19     print('Free shipping!')
20 else:
21     shipping_cost = customer_basket_weight * shipping_cost_per_kg
```

Exercise Complete! The flag is: THM{MY\_FIRST\_APP}

Python code output

Free shipping!  
Total basket cost including shipping is 101

## 7.1 Loops

On the code editor, click back on the "script.py" tab and code a loop that outputs every number from 0 to 50.

Answer:THM{L00PS\_WHILE\_FOR}

Below is some example Python code that will print the numbers from 0 to 4. In programming, 0 is often the starting number, so counting to 5 is 0 to 4 (but has 5 numbers: 0, 1, 2, 3, and 4)

```
for i in range(5):  
    print(i)
```

Answer the questions below

On the code editor, click back on the "script.py" tab and code a loop that outputs every number from 0 to 50.

THM{LOOPS\_WHILE\_FOR}

✓ Correct Answer



Task 8 Bitcoin Project Introduction to Functions

Below is some example Python code that will print the numbers from 0 to 4. In programming, 0 is often the starting number, so counting to 5 is 0 to 4 (but has 5 numbers: 0, 1, 2, 3, and 4)

```
for i in range(5):  
    print(i)
```

Answer the questions below

On the code editor, click back on the "script.py" tab and code a loop that outputs every number from 0 to 50.

THM{LOOPS\_WHILE\_FOR}

✓ Correct Answer



Task 8 Bitcoin Project Introduction to Functions

```
script.py  flag.txt  shipping.py  bitcoin.py  Code  
1 # Write your python code here  
2 i = 0  
3 while i <= 50:  
4     print(i)  
5     i = i + 1  
6  
7  
8  
9  
  
Exercise Complete! The flag is: THM{LOOPS_WHILE_FOR}  
  
Python code output  
0  
1  
2  
3  
4  
5  
6  
7  
8
```

```
script.py  flag.txt  shipping.py  bitcoin.py  Run Code  
1 # Write your python code here  
2 i = 0  
3 while i <= 50:  
4     print(i)  
5     i = i + 1  
6  
7  
8  
9  
30  
39  
40  
41  
42  
43  
44  
45  
46  
47  
48  
49  
50
```

## 8.1 Introduction to Functions

a) Once you've written the bitcoinToUSD function, use it to calculate the value of your Bitcoin in USD, and then create an if statement to determine if the value falls below \$30,000; if it does, output a message to alert you (via a print statement).

Answer: THM{BITCOIN\_INVESTOR}

which is your bitcoin value in USD (to calculate this, in the function, you times bitcoin\_amount variable by bitcoin\_value\_usd variable and return the value). The start of the function should look like this:

```
def bitcoinToUSD(bitcoin_amount, bitcoin_value_usd):
```

Once you've written the bitcoinToUSD function, use it to calculate the value of your Bitcoin in USD, and then create an if statement to determine if the value falls below \$30,000; if it does, output a message to alert you (via a print statement).

THM{BITCOIN\_INVESTOR}

✓ Correct Answer



```
script.py  flag.txt  shipping.py  bitcoin.py  Code  
19 def bitcoinToUSD(bitcoin_amount, bitcoin_value_usd):  
20     usd_value = bitcoin_amount * bitcoin_value_usd  
21     return usd_value  
22  
23 investment_in_usd = bitcoinToUSD(investment_in_bitcoin,  
24     bitcoin_to_usd)  
25 if investment_in_usd <= 30000:  
26     print("Investment below $30,000! SELL!")  
27 else:  
28     print("Investment above $30,000")  
  
Exercise Complete! The flag is: THM{BITCOIN_INVESTOR}  
  
Python code output  
Investment above $30,000
```

b) Bitcoin is now worth \$24,000. In the code editor on line 14, update the `bitcoin_to_usd` variable value to 24000 and see if your Python program recognises that your investment is below the \$30,000 threshold.

Answer:

Once you've written the `bitcoinToUSD` function, use it to calculate the value of your Bitcoin in USD, and then create an if statement to determine if the value falls below \$30,000; if it does, output a message to alert you (via a print statement).

THM{BITCOIN\_INVESTOR}

✓ Correct Answer

1 Bitcoin is now worth \$24,000. In the code editor on line 14, update the `bitcoin_to_usd` variable value to 24000 and see if your Python program recognises that your investment is below the \$30,000 threshold.

No answer needed

✓ Correct Answer

Task 9 Files

```
script.py  flag.txt  shipping.py  bitcoin.py  Code
14
15 investment_in_bitcoin = 1.2
16 bitcoin_to_usd = 24000
17
18 # 1) write a function to calculate bitcoin to usd
19 def bitcoinToUSD(bitcoin_amount, bitcoin_value_usd):
20     usd_value = bitcoin_amount * bitcoin_value_usd
21     return usd_value
22
```

Exercise Complete! The flag is: THM{BITCOIN\_INVESTOR}

Python code output

Investment below \$30,000! SELL!

## 9.1 Files

In the code editor, write Python code to read the `flag.txt` file. What is the flag in this file?

Answer: THM{FILE\_READ}

Notice we use the `close()` method after writing to a file; this closes the file so no more writing to the file (within the program) can occur.

Answer the questions below

In the code editor, write Python code to read the `flag.txt` file. What is the flag in this file?

THM{FILE\_READ}

✓ Correct Answer

Task 10 Imports

```
script.py  flag.txt  shipping.py  bitcoin.py  Run Code
1 f = open("flag.txt", "r")
2 print(f.read())
3
4
5
6
7
8
9
```

Python code output

THM{FILE\_READ}

## 10.1 Import

I was expected to read through the text and understand some of the libraries that a pentester needs to know `l.eRequest` - simple HTTP library, `Scapy` - send, sniff, dissect and forge network packets, `Pwntools` - a CTF & exploit development library.

Answer the questions below

Read the task and run the Python example code above in the code editor on the right.

No answer needed

✓ Correct Answer

Below is a screenshot of my completion of task and a link to my achievement.

[https://tryhackme.com/room/pythonbasics?utm\\_campaign=social\\_share&utm\\_medium=social&utm\\_content=share-completed-room&utm\\_source=copy&shareId=6969d963f9c9a9590800e42d](https://tryhackme.com/room/pythonbasics?utm_campaign=social_share&utm_medium=social&utm_content=share-completed-room&utm_source=copy&shareId=6969d963f9c9a9590800e42d)

